Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Photism
b. Team leader name: Khunapat Prapassornchaikul
c. Problem Statement: Media, Content & Digital Experiences. FaceFinder turns one shared Google Drive event album into one folder per person, with Gemini picking and sorting the best photos.

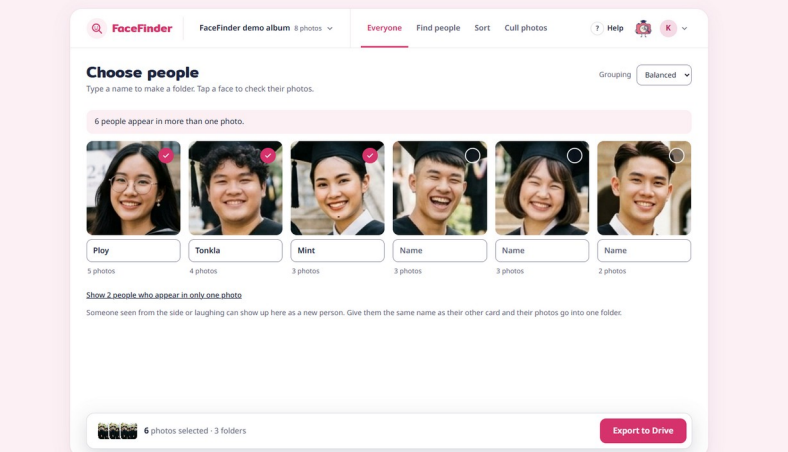
Brief about the idea

The problem

After an event, the photographer shares one Google Drive folder with hundreds of photos. Finding yourself, or sending friends their photos, means an hour of scrolling.

FaceFinder

Scan the album once. A face model groups faces into people; Gemini judges and describes each photo. Then find a person, keep the best shots, sort by description, and copy one folder per person into your Drive. The album is never changed.



The live app on the demo album. The people are made up, created with Gemini.

Opportunities

How is it different?

Event photos usually live in one shared Drive folder, with no face search and no culling. FaceFinder works on any shared Drive folder, for everyone who can open it, and gives back real Drive folders.

How does it solve the problem?

One scan per album. Name a face or upload a selfie to get that person's photos. Gemini suggests keepers with a reason, and sorts by description. One click copies it all to your Drive.

USP

Two AIs, each for its job: a face model decides who; Gemini decides how good and what is in it. Copy-only, no photo library kept, data deletable at any time. Works on a phone.

List of features offered by the solution



Everyone

Every face grouped into people. Type a name to make that person's folder.



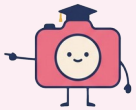
Find people + Best 10

One selfie finds a person. Best 10 picks their best shots, each with Gemini's reason.



Cull photos

Keep, Review or Rejected, with reasons: blur, eyes closed, weaker shots of a burst.



Sort by description

Describe a folder in plain words ("throwing caps"). Gemini fills it and suggests folders.



Export to Drive

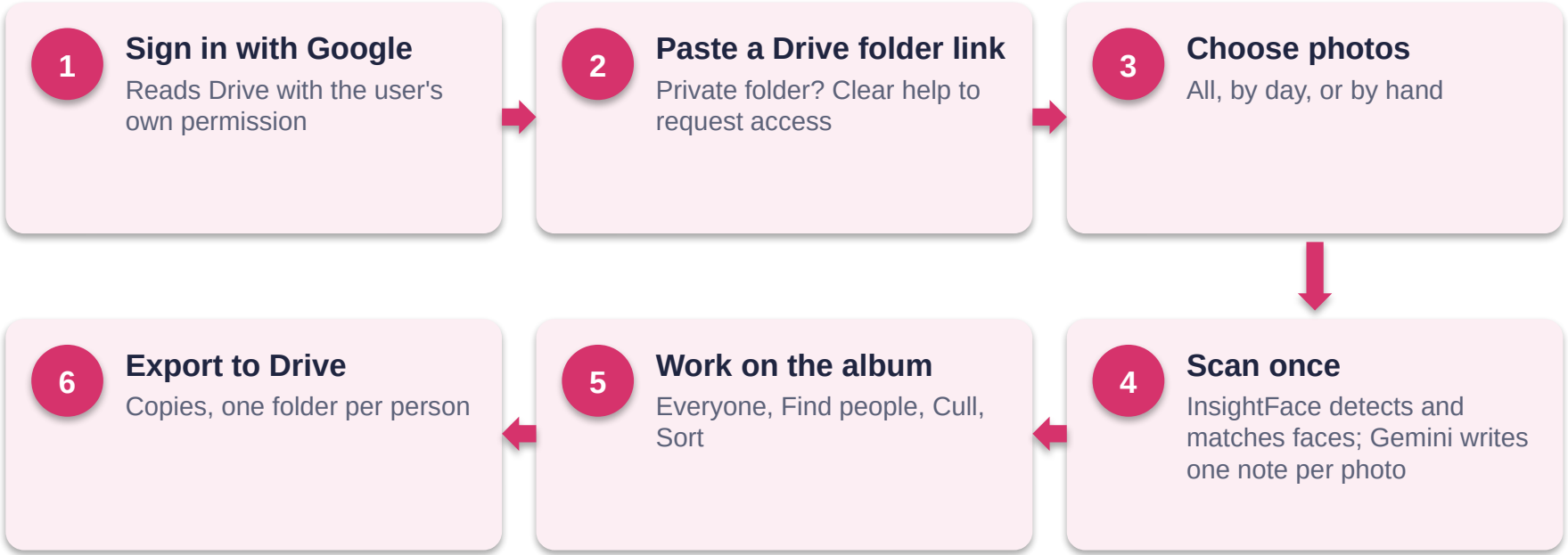
One folder per person in your own Drive. Copies are made inside Google; never duplicated.



Control and privacy

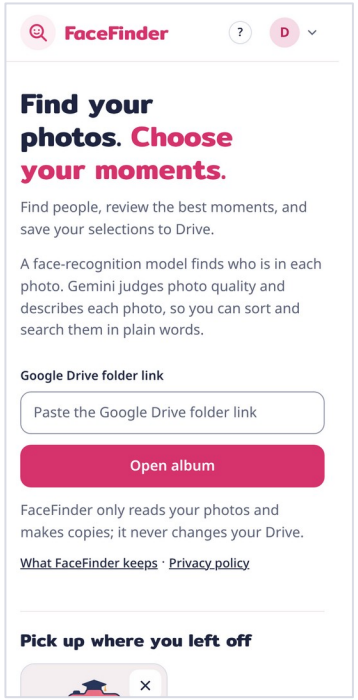
Choose photos by day or by hand, stop and resume. See what is kept, delete it any time.

Process flow diagram or Use-case diagram

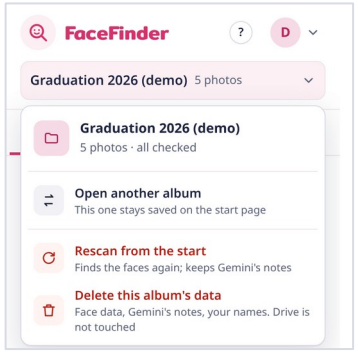


Who uses it: the photographer culls and sorts; guests find themselves; friends share each person's folder.

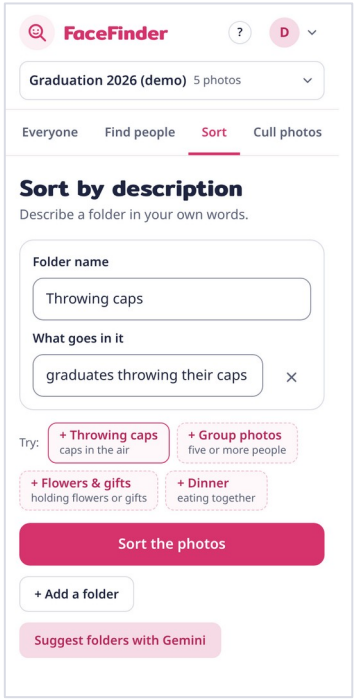
Wireframes/Mock diagrams of the proposed solution (optional)



Start page



Album menu

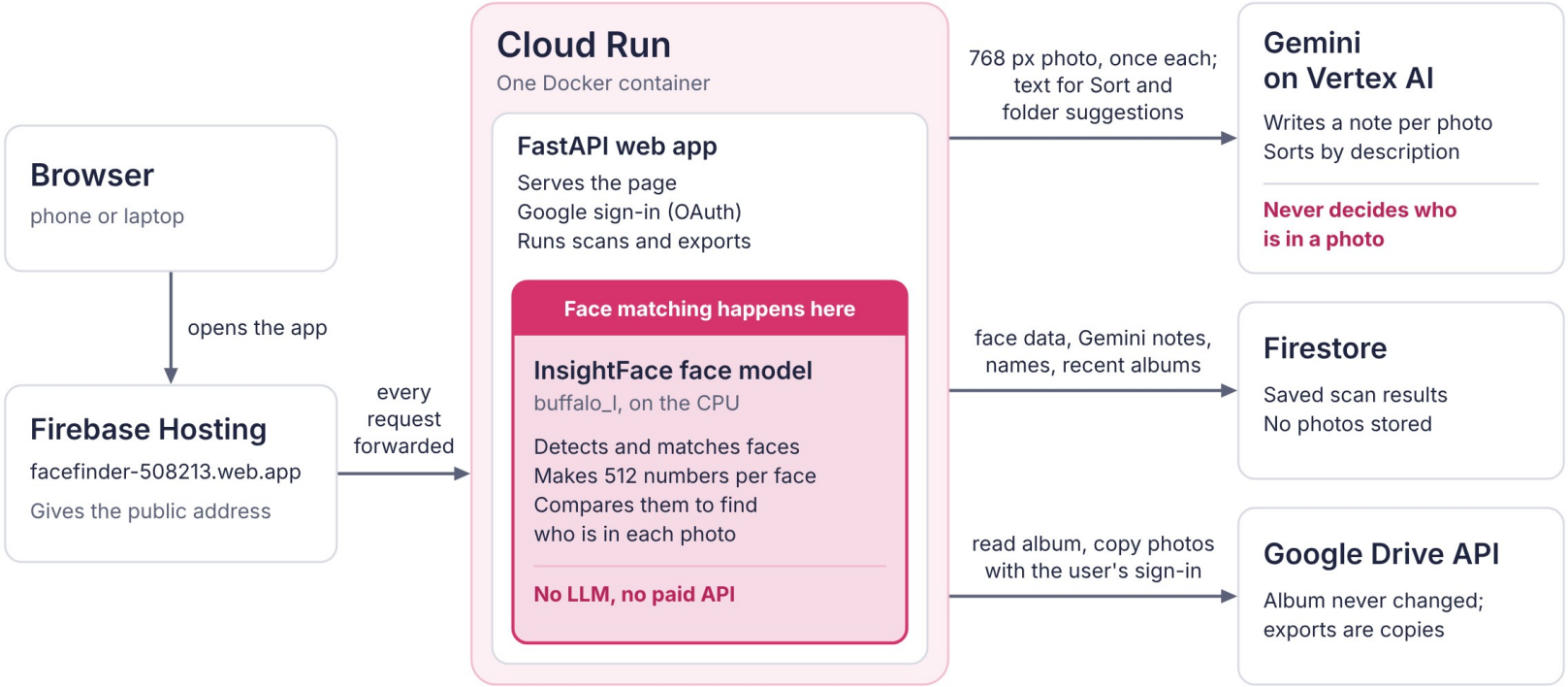


Sort by description

Phone first

Works from 320 px wide to desktop, with large touch targets and a bottom action bar. Our own pink design and camera mascot.

Architecture diagram of the proposed solution



Technologies to be used in the solution

Google Cloud

Cloud Run: runs the app and the face model

Firestore: scan results, Gemini notes, names

Firebase Hosting: the public address

Vertex AI: Gemini, no API key needed

Drive API, Google sign-in: read albums, copy photos

Secret Manager: sign-in secrets

AI models

Gemini 3.5 Flash: photo notes, sorting, folder suggestions

InsightFace buffalo_l: face detection and matching, on CPU

OpenCV: images, burst fingerprints

App and tools

Python, FastAPI: server and background jobs

HTML, CSS, JavaScript: one-file front end

Docker: one container

Tests: 90 Python checks, browser tests

Antigravity, Gemini: testing, demo photos of made-up people

Estimated implementation cost (optional)

What it costs to run

- **Cloud Run:** one 4-vCPU instance; it scales to zero when idle.
- **Gemini:** one image call per photo. Cached notes avoid repeating it for unchanged photos; Sort sends text only.
- **After the scan:** Find people, Cull and Best 10 reuse saved notes, with no new Gemini calls.
- **Still charged:** compute, Firestore storage and reads, as usual. No measured per-album cost is claimed.

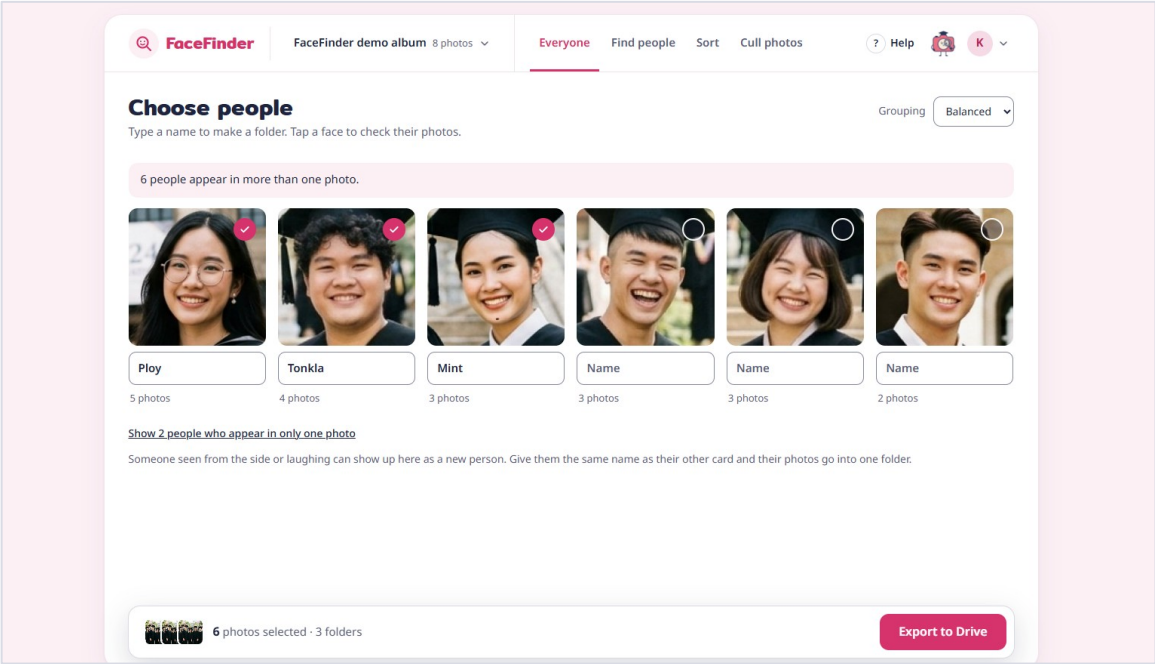
Per 1,000 photos

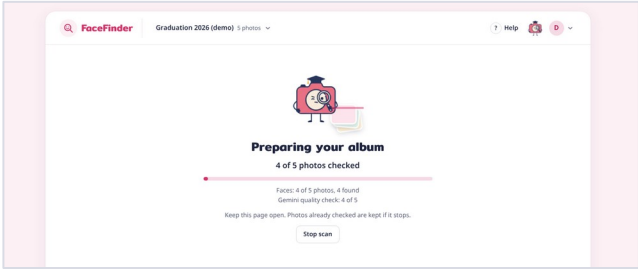
1,000
Gemini image calls, once

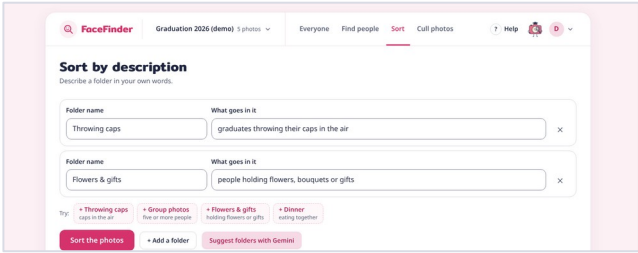
~25
text calls for each sort

0
new Gemini image calls to rescan unchanged photos

Snapshots of the prototype







Everyone on the live app: name a face to make a folder. Right: scanning, and Sort by description.

Prototype Performance report/Benchmarking

0.75–0.98
face match score for the same person, against −0.05 for a different person

40 of 40
small-face stress test: 40 faces 18 px wide in a synthetic grid

1 min 36 s
one observed live run: 37 photos scanned with Gemini notes (4 vCPU)

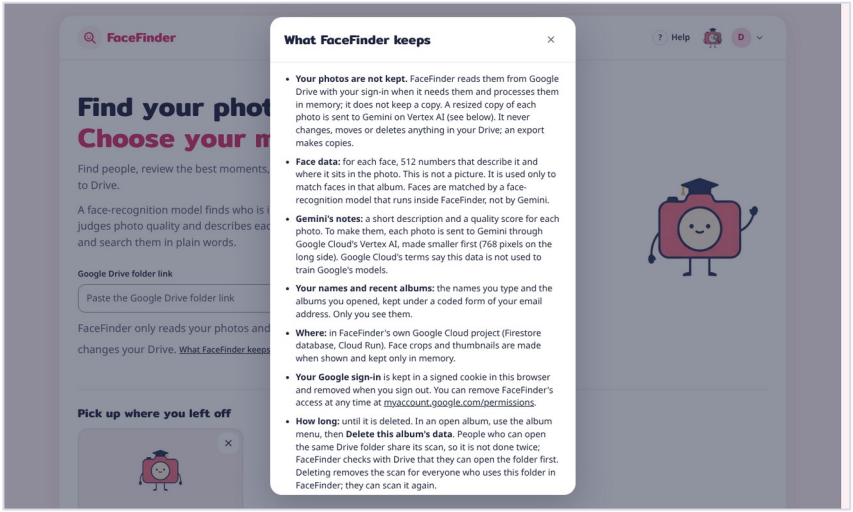
28 s
one observed live run: 31 copies exported into 5 Drive folders

16 s
one observed live run: upload a selfie, all 3 of that person's photos found

5.2 s
one observed live run: Stop ends a scan

Scores and the stress test: the repository's automatic tests. Timings: single observed runs on the live app, 17–18 October 2026.

Additional Details/Future Development (if any)



Trust today: no photo library kept, a plain list of what is kept, a public privacy policy, and "Delete this album's data".

Next steps

- **Scale:** persist job progress in Firestore, then run scans on Cloud Run jobs or Cloud Tasks.
- **Guest mode:** a QR code at the event; each guest finds their own photos.
- **Commercial use:** a commercially licensed face model.
- **Google verification:** no more "unverified app" warning.

Provide links to your:

- GitHub Public Repository: https://github.com/Khunapat/facefinder
- Demo Video Link (3 to 4 minutes): https://youtu.be/1NW303-InI0
- Final Product Link: https://facefinder-508213.web.app

For judges

Open the app and sign in with Google. Google may show "Google hasn't verified this app": press Advanced, then Go to facefinder. Open the Sample album card (made-up people created with Gemini) and try Everyone, Find people, Sort and Cull photos.

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Thank you

